COURSE SLIDES

# PL-7002 Create and Manage Automated Processes by using Power Automate

Course Code: C903

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

# Welcome & Housekeeping

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TRAINERS LISTED FOR C903

# Course Trainer Profiles

1

## Assigned Tertiary Infotech Academy Trainer

Power Platform practitioner experienced in cloud-flow design, business-process automation, data integration, monitoring, and operational handover.


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ICE-BREAKER

# Let's Know Each Other

1. Your name, organisation, and role.
2. One repeated task you would like to automate.
3. The event that should start the automation.
4. The evidence that would prove the automation worked.


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HOUSEKEEPING

# Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. Learn by doing — try every hands-on lab.

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# Access Course Materials

1. Open the C903 course page or the trainer-supplied Drive folder
2. Open the courseware links for this course
3. Download the Slides, Learner Guide and Lesson Plan
4. Open the Activities folder for all 8 labs
5. Keep the files together as you work

**Course page: https://www.tertiarycourses.com.sg/pl-7002-create-and-manage-automated-processes-by-using-power-automate.html**

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SCHEDULE

# Lesson Plan — 1 Day, 7.5 instructional hours

## Course sequence

- Day 1 — Build, govern, integrate, and operate one connected Northwind service-request automation
  - Topic 1: Get Started with Power Automate (Labs 1–2)
  - Topic 2: Build Approval Flows with Power Automate (Labs 3–4)
  - Topic 3: Build Flows to Manage User Information (Labs 5–6)
  - Topic 4: Power Automate's Deep Integration Across Multiple Data Sources (Labs 7–8)

## Day format

- Daily timing
  - 9:30 am - 6:30 pm (1-hour lunch; two 15-minute tea breaks; 7.5 instructional hours)
- Instructional time
  - 7.5 hours · 8 connected labs
- Learning arc
  - Concept → demonstrate → build → verify → discuss → recap


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WHAT YOU'LL ACHIEVE

# Learning Outcomes

1

## Build & Diagnose

Design, build, test, share, and troubleshoot cloud flows by selecting appropriate triggers, actions, recurrence settings, dynamic content, and run-history evidence.

2

## Approve & Branch

Implement conditional logic, approval patterns, run-after behaviour, and a conditional business process flow for a governed request lifecycle.

3

## Personalise & Schedule

Create user-driven flows that use explicit inputs, Microsoft 365 user information, date and time logic, and calendar actions while controlling duplicates and data exposure.

4

## Integrate & Operate

Integrate SharePoint, Power Apps, Dataverse, Dynamics 365 patterns, and SQL; then monitor, share, own, and document a reliable automated process.

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LEARNING APPROACH

# How You'll Learn

1. Trainer demonstrates the concept
2. You build it yourself in the hands-on lab
3. Verify with the lab's 'Test it' step
4. Discuss and troubleshoot together
5. Recap against the learning outcomes

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CORE CONCEPTS

# The Observable Automation Loop

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OVERVIEW

# Keep Six Design Questions Visible

1. **Event**
   What exact event or user decision creates one run?

2. **Data**
   Which system owns each field, type, and record identifier?

3. **Decision**
   Which rules are deterministic and which require a person?

4. **Action**
   What will be created, updated, sent, or returned?

5. **Evidence**
   Which run output and target record prove the outcome?

6. **Owner**
   Who monitors, recovers, changes, and reconnects the automation?


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THE BUILDING BLOCKS

# E-D-D-A-E-O Design Framework

1

## Event

Trigger type, source, timing, volume, and qualifying rule.

2

## Data

Source schema, types, required values, nulls, and correlation key.

3

## Decision

Conditions, approval outcomes, branch rules, and timeouts.

4

## Action

Connector operation, identity, target, and side effect.

5

## Evidence

Run history, target-system record, test case, and expected result.

6

## Owner

Primary and backup owner, connection, monitor, and recovery path.


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C903 OPERATING PRINCIPLE

# A flow is complete only when its trigger, result, and recovery path are observable.

Build the normal path, prove the boundary paths, and hand over the identities, evidence, and recovery actions that keep the automation operating.


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# What You Will Build

## Request layer

- SharePoint intake
- Dataverse lifecycle record
- Power Apps input contract

## Automation layer

- acknowledgement
- reminder
- approval
- calendar and integration flows

## Operations layer

- run evidence
- SQL audit row
- error record
- handover runbook

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# The Learning Arc in Every Lab

1. Open the previous checkpoint and state the expected outcome.
2. Build the bounded trigger, actions, data map, and branch.
3. Test with the supplied synthetic Request ID and inspect run evidence.
4. Save the checkpoint and record what the next lab is allowed to use.

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BEFORE A FLOW IS HANDED OVER

# The C903 Flow Release Gate

1

## Trigger

Does one qualifying event create one intended run?

2

## Data

Are types, nulls, identities, time zones, and correlation keys explicit?

3

## Logic

Have normal, false, empty, failure, timeout, and duplicate paths been exercised?

4

## Security

Are connections, permissions, data policies, and recipients appropriate?

5

## Evidence

Can a reviewer trace the test record through run history to every target outcome?

6

## Ownership

Are primary and backup owners, monitoring, reconnection, and recovery documented?


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01

TOPIC 01

# Get Started with Power Automate

Cloud-flow anatomy · automated, instant, and scheduled triggers · dynamic content · testing · sharing · run history · troubleshooting

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# Key Concepts — Get Started with Power Automate

1

## Trigger

The event or schedule that creates one flow run and supplies the initial data payload.

2

## Action

A configured operation that reads, transforms, sends, creates, or updates data after the trigger.

3

## Connection

The authenticated relationship a connector uses to act against a service under an identity.

4

## Dynamic content

A token that carries a value from a trigger or earlier action into a later field.

5

## Run history

The evidence trail for trigger input, action input and output, duration, status, and error details.

6

## Ownership

The people or service identity responsible for editing, licensing, connections, monitoring, and recovery.

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TOPIC 01 · CONCEPT

# Cloud-Flow Anatomy and Trigger Selection

## WHAT

- A cloud flow is an event-driven sequence made of one trigger followed by actions, control steps, and data transformations. Automated flows react to a service event, scheduled flows start from a recurrence, and instant flows start when a person or app invokes them. The trigger creates the run and provides the first payload; every later action consumes values from that payload or from earlier outputs.

## WHY IT MATTERS

- Trigger choice determines when work starts, which identity and data are available, how often the service is queried, and whether duplicate events are possible. Starting from the business event instead of from a favourite connector keeps the automation aligned with the process. A good design states the event, system of record, expected volume, latency, and evidence of success.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

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# How It Works — Cloud-Flow Anatomy and Trigger Selection

1. Write the business event as: When <event> occurs in <system>, automate <outcome> for <recipient>.
2. Choose automated, scheduled, or instant based on who or what controls the start.
3. Inspect the trigger schema and identify the stable record ID, timestamps, owner, and business fields.
4. Add only the actions needed to reach an observable result, then rename every card by intent.
5. Test with a known record and compare trigger input, action output, and target-system evidence.

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# In Practice — Cloud-Flow Anatomy and Trigger Selection

## WORKED EXAMPLE

- A new item in the Northwind Process Requests SharePoint list starts an automated flow.
- The flow captures Request ID, title, requester, priority, and due date, then sends an acknowledgement.
- The run is successful only when run history is green and the acknowledgement contains the same Request ID.

## DECISION GUIDE

- **Use when**
  - A repeatable event has a clear owner, stable data source, and observable downstream outcome.
  - The service connector exposes the required trigger and the running identity is permitted to use it.
- **Avoid when**
  - The event is ambiguous, the process changes on every occurrence, or a person must first judge whether work should start.
  - The design cannot identify a stable record key or would repeatedly process the same event without a duplicate guard.

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# Quality Lens — Cloud-Flow Anatomy and Trigger Selection

## Event

- Name the exact event and the system that owns it.

## Payload

- Identify required fields, types, null cases, and one stable correlation key.

## Evidence

- Define the target record, message, or run output that proves success.

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# Recurrence, Event Triggers, and Data Mapping

## WHAT

- A recurrence trigger creates runs at a configured interval, start time, and time zone. An event trigger reacts when a connected service reports that a record was created or changed. Dynamic content maps fields from the trigger or an earlier action into later actions; expressions transform those values when simple token mapping is not enough.

## WHY IT MATTERS

- Scheduled and event-driven patterns solve different timing problems. A recurrence is appropriate for periodic sweeps such as overdue-request reminders, while an event trigger is appropriate for a new request acknowledgement. Explicit data mapping prevents a common defect: a flow that runs successfully but sends the wrong field, formats a date incorrectly, or processes records outside scope.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

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# How It Works — Recurrence, Event Triggers, and Data Mapping

1. For recurrence, set frequency, interval, start time, and the Singapore time zone explicitly.
2. For an event trigger, create a fresh test event and allow for the connector's polling behaviour.
3. Use Filter Query or trigger conditions to reduce unnecessary runs when the connector supports them.
4. Map dynamic content by business meaning, not by a similar-looking display name.
5. Use Compose for intermediate values so the run history exposes the value used by later steps.

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# In Practice — Recurrence, Event Triggers, and Data Mapping

## WORKED EXAMPLE

- At 4:00 pm every day, a scheduled flow queries Open requests whose ReminderSent flag is No, then checks whether each Due Date is today in Singapore.
- A Runtime diagnostic Compose reads the supplied Diagnostic Divisor before Outlook sends the Request ID and Title reminder and SharePoint marks ReminderSent Yes.
- The designer proves one due item sends once, a repeated run sends nothing, a tomorrow item takes the false path, and a supplied zero creates a controlled runtime failure.

## DECISION GUIDE

- **Use when**
  - A periodic business rule can query a bounded set and tolerate the chosen interval.
  - An expression or Compose step makes a derived value easier to inspect and reuse.
- **Avoid when**
  - The process requires immediate response but the selected connector uses a long polling interval.
  - A broad scheduled query would retrieve an unbounded dataset or resend the same item without a sent flag.

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# Quality Lens — Recurrence, Event Triggers, and Data Mapping

## Clock

- Record time zone, start time, interval, and daylight-saving implications.

## Scope

- Filter at the source and define what happens when zero, one, or many rows return.

## Mapping

- Trace each dynamic token to its producing trigger or action.

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# Testing, Run History, Sharing, and Troubleshooting

## WHAT

- Testing executes a flow with controlled input and records the resulting run. Run history exposes the status, duration, inputs, outputs, retries, and error message for each step. Sharing can make another person a co-owner who can edit and maintain the flow, or a run-only user who can invoke an instant flow without redesigning it.

## WHY IT MATTERS

- A green designer is not proof of a reliable process. The flow must be exercised with representative data, branch boundaries, empty values, and expected failures. Troubleshooting begins with the symptom: no run indicates a trigger problem; a red step indicates an action or connection problem; an all-green run with a wrong outcome indicates data mapping or logic. Ownership and connection planning prevent a useful flow from stopping when its maker changes role.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

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# How It Works — Testing, Run History, Sharing, and Troubleshooting

1. Save, run a manual test, and capture the run URL or timestamp with the test record ID.
2. If no run exists, inspect trigger event, conditions, status, schedule, and connector polling.
3. If an action fails, open the first red action and read its inputs, outputs, status code, and connection.
4. If the run is green but wrong, compare dynamic content and expressions with the source record.
5. Share with the least privilege needed, confirm connections, and record primary and backup owners.

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# In Practice — Testing, Run History, Sharing, and Troubleshooting

## WORKED EXAMPLE

- A reminder flow saves but no run appears because the recurrence start time is tomorrow.
- After correcting the start time, the Outlook action fails with 401 because its connection expired.
- The maker fixes the connection, retests one known item, and records the successful run before handover.

## DECISION GUIDE

- **Use when**
  - A controlled test record can be traced from trigger to target without exposing production-sensitive data.
  - A team-owned business process needs documented co-owners, connection responsibilities, and monitoring.
- **Avoid when**
  - Testing would send real external messages or modify live records without a safe test boundary.
  - A flow is shared without checking whether recipients can access the underlying connections and data.

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# Quality Lens — Testing, Run History, Sharing, and Troubleshooting

## Coverage

- Test normal, boundary, empty, and deliberately failing cases.

## Diagnosis

- Locate the first divergence between expected and actual inputs or outputs.

## Continuity

- Name owners, connection identity, monitoring cadence, and recovery action.

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WHAT YOU'LL DO

# Hands-On Labs — Get Started with Power Automate

## Lab 1

- Validate the Northwind Request Intake and Create an Automated Acknowledgement

## Lab 2

- Build a Scheduled Reminder and Diagnose a Controlled Failure

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TOPIC 01 · HANDS-ON

# Validate the Northwind Request Intake and Create an Automated Acknowledgement

## LAB 1

You will inspect the pre-provisioned SharePoint list used throughout the course, verify five synthetic Northwind requests, and build an automated cloud flow that acknowledges every new request. You will rename the trigger and actions, map values deliberately, create a test request, and trace it from the SharePoint item through run history to the delivered message.

### You'll build

**A validated per-learner SharePoint list named C903 Process Requests, an automated cloud flow named C903-01-Acknowledge-New-Request, and a saved evidence record for RequestID NW-1006**

| 1 Create list | 2 Load baseline | 3 Build flow | 4 Trigger test |
|---|---|---|---|

**Tools:** SharePoint Online · Power Automate cloud flows · Office 365 Outlook


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LAB 1 · VERIFY

# Validate the Northwind Request Intake and Create an Automated Acknowledgement

## Test it

The isolated SharePoint list must contain NW-1001 through NW-1006 with the exact schema. A single new NW-1006 item must create one successful run. The trigger input, Correlation_key output, email subject, email body, and T01 evidence row must all contain NW-1006. The email must show the mapped title, priority, and due date and must not contain any unrelated list field.


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TOPIC 01 · HANDS-ON

# Build a Scheduled Reminder and Diagnose a Controlled Failure

## LAB 2

You will build a scheduled flow that finds open requests due today, sends one reminder, and marks each item so the next run does not resend it. You will insert a temporary divide-by-zero expression, use run history to find the first red action, restore the supplied divisor, retest the same expression, and share the flow with a trainer-designated co-owner for continuity.

### You'll build

**A scheduled cloud flow named C903-02-Remind-Due-Requests, evidence rows for normal, empty, duplicate, and controlled-failure cases, and a recorded primary and backup owner**

| 1 Schedule | 2 Filter | 3 Remind | 4 Break safely |
|---|---|---|---|

**Tools:** Power Automate scheduled flows · SharePoint Online · Office 365 Outlook


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LAB 2 · VERIFY

# Build a Scheduled Reminder and Diagnose a Controlled Failure

## Test it

T02 must show one reminder and ReminderSent set to Yes for NW-1002. T03 must prove repeat-run duplicate prevention, and T04 must prove the false-date branch. T05 must identify Runtime_diagnostic_check as the first red action with DiagnosticDivisor 0; T06 must resubmit successfully after the divisor is restored to 1 and set ReminderSent to Yes. The final flow must be on and list the approved primary and backup owners.

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TOPIC RECAP

# Recap - Get Started with Power Automate

1. LO1 - Build and Diagnose: Choose a trigger that matches the business event, configure actions with explicit data, test both expected and exceptional paths, and use run history to locate the first failure.

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02

TOPIC 02

# Build Approval Flows with Power Automate

Approval patterns · conditional logic · custom responses · trigger conditions · run-after paths · business process flows · conditional stages


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# Key Concepts — Build Approval Flows with Power Automate

1

## Approval

A human decision step that records an outcome, responder, comments, and completion time.

2

## Condition

A two-branch control that evaluates a Boolean comparison after a run has started.

3

## Trigger condition

An expression evaluated before the run is created, reducing unnecessary executions.

4

## Run after

A dependency setting that selects which next action runs after success, failure, skip, or timeout.

5

## Business process flow

A Dataverse-backed stage and step experience that guides a person through a consistent record lifecycle.

6

## Branching

A rule that changes the next path when data or a human decision meets defined criteria.

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# Approval Patterns and Human Decision Points

## WHAT

- Power Automate approvals combine an automated flow with a recorded human response. Start and wait for an approval pauses the run until the configured response condition is met. The approval type controls whether the first response, every response, a sequence, or custom response options complete the step. The returned outcome and comments become data for later branches.

## WHY IT MATTERS

- An approval is useful where policy requires a named person to accept, reject, or request clarification. It should not be added merely to make a flow look governed. The design must state who decides, what evidence they receive, how long they have, what each outcome means, and what happens if no one responds.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

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# How It Works — Approval Patterns and Human Decision Points

1. Define the decision, approver role, evidence, response options, timeout expectation, and audit record.
2. Choose first response, all responses, sequential, or custom responses based on the actual rule.
3. Build the approval title from a stable Request ID and keep the details concise and traceable.
4. Branch on the exact returned Outcome value and persist outcome, responder, comments, and completion time.
5. Notify the requester and update the system of record on every designed outcome.

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# In Practice — Approval Patterns and Human Decision Points

## WORKED EXAMPLE

- A high-priority Northwind request creates an approval for the operations manager.
- The approver chooses Approve, Need more information, or Reject and adds a comment.
- The flow updates SharePoint with the exact outcome and sends the requester a matching message.

## DECISION GUIDE

- **Use when**
  - A policy or risk decision needs a named responder and durable decision evidence.
  - The approver receives enough context and has access to any linked supporting record.
- **Avoid when**
  - The decision can be expressed safely as a deterministic rule with no human judgment.
  - The flow has no designed timeout, escalation, cancellation, or ownership path for unanswered requests.

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# Quality Lens — Approval Patterns and Human Decision Points

## Decision

- State one question and define every allowed response.

## Evidence

- Include only the facts and secure links the approver needs.

## Closure

- Persist outcome and notify the correct people on every path.

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# Conditional Logic, Trigger Conditions, and Run-After Paths

## WHAT

- A Condition action branches after a run begins. A trigger condition is evaluated at the trigger and can prevent an unwanted run from being created. Configure run after changes dependency behaviour so an action can execute after success, failure, skip, or timeout. Together these controls separate business rules from technical recovery behaviour.

## WHY IT MATTERS

- Conditions decide what the process should do with valid data; run-after settings decide what the flow should do when a step does not complete normally. Mixing them produces brittle designs. Trigger conditions can reduce noise and request consumption, but an incorrect expression can silently suppress all runs, so the expression and a known qualifying record must be tested together.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

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# How It Works — Conditional Logic, Trigger Conditions, and Run-After Paths

1. Write each business rule in plain language with field, operator, value, and null behaviour.
2. Use a Condition after the trigger when both qualifying and non-qualifying runs need recorded outcomes.
3. Use a trigger condition when non-qualifying events should create no run and that absence is acceptable.
4. Group critical actions in a Try Scope and configure Catch to run after failure, timeout, or skip.
5. Record the error, Request ID, failed step, and recovery owner before sending a controlled alert.

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# In Practice — Conditional Logic, Trigger Conditions, and Run-After Paths

## WORKED EXAMPLE

- Only High-priority requests need human approval; other priorities continue automatically.
- A Try Scope creates the approval and updates SharePoint. Catch runs only if Try fails or times out.
- The Catch path writes an Automation Error status and sends one internal notification with the Request ID.

## DECISION GUIDE

- **Use when**
  - Branch rules are explicit and test data can exercise both true and false paths.
  - A technical failure must create a controlled record and alert instead of silently stopping the process.
- **Avoid when**
  - A trigger condition would hide events that operations still needs to count or explain.
  - Configure run after is used to ignore failure without recording what failed and who will recover it.

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# Quality Lens — Conditional Logic, Trigger Conditions, and Run-After Paths

## Rule

- Document field type, operator, comparison value, and null case.

## Path

- Name branches by business meaning instead of Yes and No.

## Recovery

- Capture error context once and route it to a named owner.

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# Business Process Flows and Conditional Stages

## WHAT

- A business process flow is a Dataverse-backed, stage-based user experience attached to one or more tables. It guides users through defined stages and data steps while the underlying model-driven app stores the record. Conditional branching can direct a record to different stages based on field values. A business process flow guides interactive work; a cloud flow automates operations around records.

## WHY IT MATTERS

- Some processes require people to see the current stage, complete required data, and move the record forward consistently. A cloud flow alone is not a user guidance surface. Combining the two patterns lets the business process flow govern stage progression while cloud flows notify, approve, integrate, and monitor around the record.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

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# How It Works — Business Process Flows and Conditional Stages

1. Model the lifecycle as a small number of named stages with an owner and exit evidence for each stage.
2. Create the Dataverse table and required columns before designing the stage path.
3. Add data steps to stages and mark only genuinely required information as required.
4. Add a branch rule on a stable field such as Priority or Approval Required, then validate and activate.
5. Test one record through each branch and confirm both the active stage and the stored field values.

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# In Practice — Business Process Flows and Conditional Stages

## WORKED EXAMPLE

- Northwind uses Intake, Review, Approval, Fulfilment, and Closure stages for automation requests.
- A high-priority record enters Approval after Review; a standard request moves directly to Fulfilment.
- The record ID remains constant while cloud flows update status and notify the request owner.

## DECISION GUIDE

- **Use when**
  - People need a visible stage path and consistent data capture on a Dataverse record.
  - The organisation has an environment, table, app, roles, and licensing suitable for the process.
- **Avoid when**
  - The process is entirely system-to-system with no need for an interactive stage experience.
  - A complex diagram is being reproduced without simplifying stages, ownership, and exit criteria first.

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# Quality Lens — Business Process Flows and Conditional Stages

## Stage

- Each stage has a clear purpose, owner, required data, and exit condition.

## Branch

- Branch rules use stable values and every path reaches a defined end.

## Integration

- Cloud flows reference the same Dataverse record ID and do not duplicate process state.

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WHAT YOU'LL DO

# Hands-On Labs — Build Approval Flows with Power Automate

## Lab 3

- Build a Priority Approval with Complete Outcome and Recovery Paths

## Lab 4

- Create and Traverse a Conditional Dataverse Business Process Flow

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TOPIC 02 · HANDS-ON

# Build a Priority Approval with Complete Outcome and Recovery Paths

## LAB 3

You will build an automated flow for new SharePoint requests. High priority creates an approval with Approve, Need more information, and Reject responses; other priorities continue without an approval. The flow stores outcome, comments, and status on the same request. A one-minute training timeout exercises Catch, which filters the scope result down to action name, status, error code, and message.

### You'll build

**An automated cloud flow named C903-03-Approve-High-Priority-Request with explicit non-High actions, three approval cases, a default case, serial Try/Catch/Finally scopes, and five path tests**

| 1 Qualify | 2 Request decision | 3 Route every outcome | 4 Filter errors |
|---|---|---|---|

**Tools:** SharePoint Online · Power Automate Approvals · Office 365 Outlook · conditions and control scopes


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LAB 3 · VERIFY

# Build a Priority Approval with Complete Outcome and Recovery Paths

## Test it

T07-T09 must each create one approval, store the exact outcome and supplied comment, update the expected status, and send one matching decision message. T10 must create no approval and must take the explicit non-High update and message path. T11 must time out, run Catch, store Automation Error, and send only the filtered action name, status, code, and message. Finally must run after both successful and skipped Catch outcomes. Error_code and Error_message must both be nonblank in T11 evidence; use the documented fallback text only when the connector supplies no structured code or message.

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TOPIC 02 · HANDS-ON

# Create and Traverse a Conditional Dataverse Business Process Flow

## LAB 4

The trainer has pre-provisioned the solution, table, alternate key, main form, active view, and app shell so the timed lab focuses on process design. You will verify the model, create a five-stage business process flow, add a High-priority branch, activate it, and advance NW-1007 and NW-1008 through every expected stage.

### You'll build

**An active C903 Request Lifecycle business process flow in the C903 Northwind Automation solution, published in C903 Request Manager, with complete High and Normal path evidence**

| 1 Verify model | 2 Design stages | 3 Branch priority | 4 Publish process |
|---|---|---|---|

**Tools:** Power Apps maker portal · Microsoft Dataverse · business process flows · model-driven apps


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LAB 4 · VERIFY

# Create and Traverse a Conditional Dataverse Business Process Flow

## Test it

The solution must contain the active five-stage C903 Request Lifecycle with one High-priority branch and an active Request ID alternate key. T12 must show all five stages for NW-1007 and T13 must show the four-stage default path for NW-1008. Both records must reach Closure and end Closed. The correlation register must use the supplied header and store the SharePoint and Dataverse IDs for both RequestIDs.

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TOPIC RECAP

# Recap - Build Approval Flows with Power Automate

1. LO2 - Approve and Branch: Use an approval only where a person must decide, branch on the returned outcome, configure failure paths deliberately, and distinguish a cloud flow from a stage-based business process flow.

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03

TOPIC 03

# Build Flows to Manage User Information

Instant flow inputs · Office 365 Users · location and date values · time-zone conversion · calendar events · privacy · duplicate prevention

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# Key Concepts — Build Flows to Manage User Information

1

## Instant input

A typed value supplied by the person or app that invokes an instant cloud flow.

2

## User profile

Directory information returned by the Microsoft 365 Users connection for an authorised identity.

3

## Expression

A formula that transforms, validates, combines, or defaults values during a run.

4

## Time zone

The regional offset and daylight rule needed to interpret a local date and time correctly.

5

## Idempotency

The property that repeating the same request does not create an unintended duplicate outcome.

6

## Minimum data

The smallest set of user information needed to achieve the declared automation purpose.

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# Instant Flows and User-Input Contracts

## WHAT

- An instant cloud flow starts when a person selects a button or when Power Apps invokes the flow. The trigger can define typed inputs such as text, number, Boolean, email, date, or file. Those inputs form a contract between the user interface and the automation: names, required values, allowed values, defaults, and validation must be explicit.

## WHY IT MATTERS

- User-driven automation is appropriate when a person decides when work should begin and supplies context that no system event contains. Poor input design moves ambiguity into later actions. A short, typed input contract reduces errors, makes test cases repeatable, and avoids collecting unnecessary personal data.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

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# How It Works — Instant Flows and User-Input Contracts

1. Define the user decision and the minimum fields the flow needs to act.
2. Name inputs with business terms and include format guidance in the prompt or app.
3. Validate required values and allowed options before calling downstream connectors.
4. Create a correlation key from the request ID and invocation time or another stable input.
5. Return a concise confirmation containing the correlation key and next action.

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# In Practice — Instant Flows and User-Input Contracts

## WORKED EXAMPLE

- A coordinator runs an instant flow for Request NW-1007 and enters a follow-up date and work mode.
- The flow retrieves the coordinator's profile and selects an office or remote message branch.
- The confirmation contains the Request ID, selected date, and resolved location without exposing extra profile fields or promising a request link.

## DECISION GUIDE

- **Use when**
  - A person must choose the start moment or provide a small amount of current context.
  - The flow can validate input before creating messages, calendar events, or records.
- **Avoid when**
  - The process should start automatically from a reliable source-system event.
  - The trigger asks for sensitive or irrelevant personal information that the process does not need.

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# Quality Lens — Instant Flows and User-Input Contracts

## Contract

- Name every input, type, required state, allowed value, and default.

## Validation

- Stop early with a useful message when data is missing or malformed.

## Confirmation

- Return the correlation key and the exact outcome created.

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TOPIC 03 · CONCEPT

# Profile, Location, Date, and Time Logic

## WHAT

- The Office 365 Users connector can retrieve profile properties for the signed-in or named organisational user. Date and time values arrive in specific formats and time zones; expressions such as formatDateTime, addDays, coalesce, and convertTimeZone transform them. Location may come from an explicit input or an authorised directory property and should be used only for the declared purpose.

## WHY IT MATTERS

- Personalised flows fail when a display name is used where an email is required, an optional profile field is null, or an unlabelled timestamp is interpreted in the wrong time zone. Explicit types, null handling, and time-zone conversion make the result predictable. Minimum-data design reduces privacy and maintenance risk by retrieving only what later actions use.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.


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# How It Works — Profile, Location, Date, and Time Logic

1. Use the signed-in user's email or object identity as the profile lookup key.
2. Inspect which profile fields can be empty and apply coalesce only with a truthful default.
3. Keep UTC for system comparison and convert to Singapore Standard Time for local communication.
4. Format dates only at the presentation boundary; preserve the original machine-readable value.
5. Do not infer sensitive characteristics from office, department, manager, or other profile fields.

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# In Practice — Profile, Location, Date, and Time Logic

## WORKED EXAMPLE

- The flow reads only Mail, Display Name, and Office Location for the invoking user.
- If Office Location is empty, it uses the explicit Work Mode input instead of inventing a location.
- The date-only Follow-up Date input is formatted as dd MMM yyyy at the confirmation boundary.

## DECISION GUIDE

- **Use when**
  - A directory property is necessary for routing or communication and the user is authorised to retrieve it.
  - The design records which time zone each timestamp uses and tests a known date.
- **Avoid when**
  - A profile field is being collected for convenience but has no action tied to it.
  - A missing value would be silently replaced with a misleading default.

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# Quality Lens — Profile, Location, Date, and Time Logic

## Identity

- Use a stable email or object ID instead of a display name.

## Time

- Label UTC and local values and convert with an explicit source and destination zone.

## Privacy

- Retrieve, expose, and retain only the properties required by the flow.

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TOPIC 03 · CONCEPT

# Calendar Actions and Duplicate Prevention

## WHAT

- Calendar connector actions create, update, or retrieve events under an authenticated mailbox identity. A reliable calendar flow defines subject, start, end, time zone, attendees, body, and availability. Idempotency prevents the same business request from creating multiple events when a user retries or a connector repeats an operation.

## WHY IT MATTERS

- Calendar automation has visible consequences for the user and invited attendees. An incorrect time zone, duration, recipient, or retry can block the wrong time or send duplicate invitations. A deterministic category or correlation key lets the flow search for an existing event before creating a new one and supports later update or cancellation.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

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# How It Works — Calendar Actions and Duplicate Prevention

1. Derive a stable event key such as C903-NW-1007-FOLLOWUP from the request ID and event purpose.
2. Calculate end time from the validated start and bounded duration.
3. Search the target time window or maintain the created Event ID on the request record.
4. If an event already exists, update it or return an Already exists result instead of creating another.
5. Confirm subject, time zone, start, end, attendees, and saved Event ID in the run evidence.

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# In Practice — Calendar Actions and Duplicate Prevention

## WORKED EXAMPLE

- A coordinator requests a 45-minute follow-up block for NW-1007 at 3:00 pm Singapore time.
- The flow checks the stored CalendarEventID before Create event, persists the returned ID, then uses a bounded category search for C903-NW-1007-FOLLOWUP as count evidence.
- A second invocation detects the stored ID, returns it in the confirmation, and creates no duplicate.

## DECISION GUIDE

- **Use when**
  - The requester has authority to create the event and the subject and attendees are appropriate.
  - The process can persist or search a stable key and has a clear update or cancellation rule.
- **Avoid when**
  - The automation would invite external attendees or expose request details without an explicit release decision.
  - The design relies only on event subject text and cannot distinguish retries from genuinely new events.

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# Quality Lens — Calendar Actions and Duplicate Prevention

## Time

- Validate start, duration, time zone, and working-hours boundary.

## Key

- Persist a stable Request ID to Event ID relationship.

## Impact

- Check attendees, sensitivity, availability, and notification behaviour.

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WHAT YOU'LL DO

# Hands-On Labs — Build Flows to Manage User Information

## Lab 5

- Create a Validated User-Driven Follow-Up Flow

## Lab 6

- Block Calendar Time with Validation and Duplicate Prevention

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TOPIC 03 · HANDS-ON

# Create a Validated User-Driven Follow-Up Flow

## LAB 5

You will create an instant flow with RequestID, Follow-up Date, Work Mode, and Use Directory Location inputs. The flow normalises and validates Work Mode before reading profile data, requires exactly one SharePoint request, retrieves only Display Name, Mail, and Office Location, and sets one ResolvedLocation variable in both location branches.

### You'll build

**An instant cloud flow named C903-05-Create-User-Follow-Up with a four-field input contract, invalid-input termination, exactly-one lookup, minimum profile retrieval, and one branch-safe variable**

| 1 Define inputs | 2 Validate early | 3 Read minimum profile | 4 Set one location |
|---|---|---|---|

**Tools:** Power Automate instant flows · SharePoint Online · Office 365 Users · Office 365 Outlook


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LAB 5 · VERIFY

# Create a Validated User-Driven Follow-Up Flow

## Test it

The trigger must have four required inputs. T14 and T15 must find exactly one request and consume the same ResolvedLocation variable after the location Condition. T16 must terminate before profile retrieval and send no email. The two valid messages may expose only Display Name, Mail, resolved location, synthetic request fields, and the formatted date. No flow logic is edited between tests.


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TOPIC 03 · HANDS-ON

# Block Calendar Time with Validation and Duplicate Prevention

## LAB 6

You will configure an instant calendar flow with a Try_parse_time scope and a separate Schedule_calendar scope. The flow requires exactly one SharePoint request, parses an ISO local start at runtime, enforces a 15-to-120-minute future window, and creates an Outlook event only when CalendarEventID is empty. Catch reports an invalid date format, while explicit Terminate actions close invalid count, range, and past-time cases.

**You'll build**

**An instant cloud flow named C903-06-Block-Follow-Up-Time, one calendar event for NW-1007, a persisted Event ID, and evidence for six validation and duplicate cases**

| 1 Validate request | 2 Parse safely | 3 Check range | 4 Create or reuse |
|---|---|---|---|

**Tools:** Power Automate instant flows · SharePoint Online · Office 365 Outlook Calendar · control scopes

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LAB 6 · VERIFY

# Block Calendar Time with Validation and Duplicate Prevention

## Test it

T17 must create one 45-minute Busy event and persist its Event ID. T18 must reuse the same ID and the bounded Get calendar view plus category Filter array must return exactly one event. The existing NW-1007 correlation-register row must remain unique, preserve all ten columns, and carry the same CalendarEventID shown by T17/T18, SharePoint, and Outlook. T19-T22 must terminate on the stated validation path and create no event. Create event (V4) must receive date-time strings without Z and a separate UTC Time zone value. No attendees or non-synthetic request details are added. Before class, the trainer performs the bounded connector-failure acceptance check in the provisioning checklist; it must identify the leaf action and produce nonblank Error_code and Error_message (documented fallback text is acceptable only when the connector returns no structured detail). Learners do not induce this failure.

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TOPIC RECAP

# Recap - Build Flows to Manage User Information

1. LO3 - Personalise and Schedule: Collect the minimum user input, resolve profile and date values predictably, convert time zones, and use a stable identifier to avoid creating duplicate calendar events.

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04

TOPIC 04

# Power Automate's Deep Integration Across Multiple Data Sources

Connections and connectors · SharePoint · Power Apps · Dataverse and Dynamics 365 patterns · SQL · gateways · monitoring · ownership · operational handover

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# Key Concepts — Power Automate's Deep Integration Across Multiple Data Sources

1

## Connector

A packaged interface exposing triggers and actions for a service or data source.

2

## System of record

The authoritative platform that owns a business entity and governs its current state.

3

## Dataverse

The Power Platform data service for tables, relationships, security roles, and solution-aware components.

4

## Gateway

A managed bridge that lets cloud services reach supported on-premises data sources.

5

## Correlation ID

One identifier carried across systems so records and run evidence can be reconciled.

6

## Operations runbook

A maintained guide for ownership, dependencies, monitoring, common failures, and recovery actions.

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# Connectors, Connections, Identity, and Policy

## WHAT

- A connector defines available operations for a service; a connection authenticates those operations under a user or service identity. Standard and premium classifications affect licensing. Data policies can restrict which connectors may be used together, and an on-premises data gateway can provide a managed route to supported internal sources such as SQL Server.

## WHY IT MATTERS

- A multi-source flow is only as reliable as its identities, permissions, policies, and connection lifecycle. A maker may be allowed to read SharePoint but not write SQL, or a connector combination may be blocked. Design therefore begins with a connection inventory that records system, operation, identity, owner, licence, policy group, environment, and renewal responsibility.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

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# How It Works — Connectors, Connections, Identity, and Policy

1. List each system of record, required trigger or action, data classification, and expected volume.
2. Select the connector and confirm whether the required operation is standard or premium.
3. Choose a maintainable identity and grant only the permissions the action needs.
4. Check environment policy and whether a gateway, firewall rule, or administrator setup is required.
5. Record connection owner, recovery contact, and which flows depend on the connection.

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# In Practice — Connectors, Connections, Identity, and Policy

## WORKED EXAMPLE

- The Northwind solution reads a SharePoint request, writes a Dataverse row, and inserts one SQL audit record.
- The training SQL connection is premium and is owned by a managed training identity through an approved gateway.
- The runbook records the connection name, owner, table permission, gateway, and reconnection procedure.

## DECISION GUIDE

- **Use when**
  - The integration inventory is approved and each identity has a documented operational owner.
  - The connector and gateway can meet data location, policy, licensing, latency, and volume needs.
- **Avoid when**
  - A personal maker connection would become an undocumented production dependency.
  - The design attempts to bypass data policy, licensing, or network controls.

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# Quality Lens — Connectors, Connections, Identity, and Policy

## Authority

- Name the system that owns each field and status.

## Identity

- Record who authenticates, what they can do, and who maintains the connection.

## Policy

- Check connector classification, environment, licence, gateway, and data rules.

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TOPIC 04 · CONCEPT

# SharePoint, Power Apps, Dataverse, Dynamics 365, and SQL Patterns

## WHAT

- SharePoint lists support collaborative records and document-centred processes. Power Apps can collect validated user input and invoke a flow. Dataverse provides relational tables, security, and solution-aware components used by model-driven apps and Dynamics 365. SQL Server provides relational queries and commands, often through premium connections and, for on-premises systems, a gateway.

## WHY IT MATTERS

- Integration should not make every system a competing source of truth. The design assigns ownership: SharePoint holds the original intake, Dataverse holds governed lifecycle state, and SQL holds an append-only integration audit. Power Apps provides the interaction surface. The same Request ID and explicit field map allow the flow to reconcile data without confusing display names, option values, or data types.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

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# How It Works — SharePoint, Power Apps, Dataverse, Dynamics 365, and SQL Patterns

1. Declare the system of record for request, stage, decision, calendar event, and audit evidence.
2. Map source field, type, required state, transformation, target field, and null behaviour.
3. Carry Request ID as the correlation key through Power Apps input, SharePoint, Dataverse, and SQL.
4. Use parameterised connector fields or stored procedures; never concatenate untrusted input into SQL text.
5. Test create, update, duplicate, null, and permission-denied cases with synthetic data.

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# In Practice — SharePoint, Power Apps, Dataverse, Dynamics 365, and SQL Patterns

## WORKED EXAMPLE

- Power Apps sends Request ID and a validated review note to a flow.
- Flow 07A requires exactly one SharePoint item, creates or returns the keyed Dataverse row, and responds with message, Dataverse row ID, and Created/Existing state.
- The Dataverse Added event starts flow 07B, which validates the source again, inserts one SQL audit row, writes identifiers to SharePoint, and is reconciled later with the Created, Added, Existing, and any recovery-run timestamps in their defined columns.

## DECISION GUIDE

- **Use when**
  - Each system has a distinct role and the field map preserves identifiers and types.
  - The flow uses approved connections and synthetic or authorised data during testing.
- **Avoid when**
  - The flow writes the same status independently in several systems without an authority rule.
  - SQL text is built by concatenating user input or a shared connection has broader rights than required.

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# Quality Lens — SharePoint, Power Apps, Dataverse, Dynamics 365, and SQL Patterns

## Map

- Document source, type, transform, target, required state, and null behaviour.

## Key

- Carry one stable Request ID and persist downstream IDs returned by each system.

## Security

- Use parameterised actions, least privilege, and approved connection ownership.

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TOPIC 04 · CONCEPT

# Monitoring, Ownership, Reliability, and Handover

## WHAT

- Operational management keeps a flow observable and maintainable after it is built. Run history shows individual executions; analytics and alerts reveal trends; owners and co-owners maintain logic and connections; solutions organise related components for controlled movement. A runbook records purpose, dependencies, service targets, monitoring cadence, common failures, recovery, and change ownership.

## WHY IT MATTERS

- A successful classroom run proves only one path at one moment. Production reliability requires ongoing evidence: whether triggers are arriving, actions are succeeding, duration and failure rates are acceptable, connections are healthy, and unresolved errors have owners. A tested Catch path and a concise runbook reduce recovery time and keep the process independent of one maker's memory.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

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# How It Works — Monitoring, Ownership, Reliability, and Handover

1. Define service expectations: schedule, volume, acceptable latency, failure threshold, and retention needs.
2. Monitor the trigger count, success rate, duration, connector errors, and business completion record.
3. Use Try, Catch, and Finally scopes with run-after settings and one structured error record.
4. Assign primary and backup owners and confirm licensing, connection, and underlying-data access.
5. Complete a runbook and rehearse one failure, recovery, retest, and communication sequence.

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# In Practice — Monitoring, Ownership, Reliability, and Handover

## WORKED EXAMPLE

- A synthetic Diagnostic Divisor makes a Compose action fail at runtime before the SQL insert.
- Catch sets SharePoint Status to Automation Error and sends an internal alert containing only Request ID, failed action, status, error code, message, UTC time, and owner.
- After Finally records completion, a Boolean failure flag terminates the handled run Failed. The owner removes the diagnostic, resubmits that same failed 07B run, and verifies one Dataverse row and one SQL audit row, not duplicates.

## DECISION GUIDE

- **Use when**
  - A business process needs repeatable monitoring and more than one person can perform recovery.
  - The team can define an acceptable service level and retain enough run evidence to diagnose defects.
- **Avoid when**
  - The flow is handed over without access checks, owner acceptance, or a tested recovery path.
  - Alerts send raw connector payloads or personal data to recipients who do not need them.

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# Quality Lens — Monitoring, Ownership, Reliability, and Handover

## Observe

- Track trigger arrival, success, duration, business completion, and connection health.

## Recover

- Record the first failure, owner, safe retry rule, and post-recovery verification.

## Own

- Name primary and backup owners, review cadence, and change process.

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WHAT YOU'LL DO

# Hands-On Labs — Power Automate's Deep Integration Across Multiple Data Sources

## Lab 7

- Build an Idempotent Power Apps, SharePoint, Dataverse, and SQL Integration

## Lab 8

- Recover the Same Integration Run and Complete Operational Handover

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TOPIC 04 · HANDS-ON

# Build an Idempotent Power Apps, SharePoint, Dataverse, and SQL Integration

## LAB 7

You will use per-learner SharePoint, Dataverse, canvas-app, and SQL resources prepared by the trainer. The Power Apps flow rejects zero or multiple SharePoint matches and returns an existing Dataverse row instead of creating a duplicate. A Dataverse-added flow inserts one SQL audit row and writes the target identifiers back to SharePoint. The app disables its button while a submission is running.

### You'll build

**Two solution-aware guarded cloud flows, a bound C903 Request Launcher app, exactly one Dataverse and SQL row for NW-1009, and a complete supplied-header correlation record**

1 Preflight isolation

2 Build guarded app flow

3 Build event flow

4 Disable double submit

**Tools:** Power Apps · SharePoint Online · Microsoft Dataverse · SQL Server connector · Power Automate solutions

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LAB 7 · VERIFY

# Build an Idempotent Power Apps, SharePoint, Dataverse, and SQL Integration

## Test it

T23 must show one SharePoint NW-1009, one Dataverse row by the active Request ID alternate key, and one SQL row by the count query. The first app response is Created and any repeated submission is Existing with the same Dataverse GUID; no second Dataverse trigger or SQL row occurs. SharePoint DataverseRowID and SQLAuditID must match the targets. Evidence must show exactly three relevant runs (07A Created, 07B Added, 07A Existing), the app button must disable while varSubmitting is true, and the correlation register must use all ten supplied columns. The SQL count is recorded only after the NW-1009 07B run reaches terminal Succeeded.


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TOPIC 04 · HANDS-ON

# Recover the Same Integration Run and Complete Operational Handover

## LAB 8

You will wrap 07B in serial Try, Catch, and Finally scopes. A temporary Compose divides by the runtime Diagnostic Divisor, so NW-1099 value 0 fails only after the flow is saved and triggered. Catch filters error details and records Automation Error. You will remove the temporary action, resubmit the same failed 07B run, prove one Dataverse and one SQL row for NW-1099, and then run NW-1010 as an independent regression.

**You'll build**

**A final 07B flow with tested serial Try/Catch/Finally behaviour, same-record recovery evidence for NW-1099, NW-1010 regression evidence, approved co-owners, and a Reviewed operations runbook**

| 1 Wrap scopes | 2 Fail NW-1099 | 3 Correct and resubmit | 4 Run regression |
|---|---|---|---|

**Tools:** Power Automate monitoring · run-after scopes · SharePoint · Dataverse · SQL · solution ownership

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LAB 8 · VERIFY

# Recover the Same Integration Run and Complete Operational Handover

## Test it

T24A must show a runtime, not save-time, failure for NW-1099; filtered Catch evidence; and counts 1/1/0. The post-Finally failure flag must make the overall T24A run Failed and therefore resubmittable. T24B must be a resubmission of that same failed 07B run and end with counts 1/1/1, reconciled IDs, Try success, Catch skipped, and Finally success. T25 must independently show 1/1/1 for NW-1010. The final saved 07B must contain no diagnostic action, both flows must have approved primary and backup owners, and the runbook must document exact resubmission, duplicate, timeout, invalid-input, 401/403, and connection-repair paths. The correlation register must include one complete row each for NW-1099 and NW-1010 with initial and recovery 07B timestamps placed in their defined columns. T24A must identify C903_Runtime_Diagnostic as the leaf Error_action and store nonblank Error_code and Error_message; documented fallback text is acceptable only when the action returns no structured detail.

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TOPIC RECAP

# Recap - Power Automate's Deep Integration Across Multiple Data Sources

1. LO4 - Integrate and Operate: Select connectors by system of record, identity, licensing, policy, and latency; preserve a shared correlation ID; then hand over the flow with owners, monitoring checks, and a tested recovery path.

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WRAP-UP

# Course Summary & Next Steps

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LEARNING OUTCOMES

# What You Achieved

1

## Build & Diagnose

Design, build, test, share, and troubleshoot cloud flows by selecting appropriate triggers, actions, recurrence settings, dynamic content, and run-history evidence.

2

## Approve & Branch

Implement conditional logic, approval patterns, run-after behaviour, and a conditional business process flow for a governed request lifecycle.

3

## Personalise & Schedule

Create user-driven flows that use explicit inputs, Microsoft 365 user information, date and time logic, and calendar actions while controlling duplicates and data exposure.

4

## Integrate & Operate

Integrate SharePoint, Power Apps, Dataverse, Dynamics 365 patterns, and SQL; then monitor, share, own, and document a reliable automated process.

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NEXT STEPS

# Continue Building Observable Automations

1. Apply E-D-D-A-E-O to one small, authorised business process.
2. Carry one stable correlation key through every system and evidence record.
3. Test the normal, empty, false, failure, timeout, and duplicate paths.
4. Review current Microsoft guidance whenever connectors or designer controls change.

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# Keep Practising After the Course

1. Re-run the labs on your own machine — repetition is what makes the skills stick.
2. Apply the techniques to a real process or project in your own organisation.
3. Keep your downloaded C903 course files and use the lab checkpoints to restart safely.
4. Explore the follow-on courses at www.tertiarycourses.com.sg.

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PL-7002 POWER AUTOMATE · C903

# Thank You!

You can now design, build, integrate, test, share, monitor, and hand over Power Automate processes with observable results and controlled recovery.


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